121. **(c) Iso-octane** — Iso-octane has the highest octane number (assigned value 100). **(c)**

122. **b) n-Propane** — In Wurtz reaction only coupled products (C_2-C_2, C_3-C_3, and C_2-C_3) are formed; propane (C_3) cannot be produced.

123. (a) Fractional distillation is based on the difference in the boiling point of different components.

124. (d) Tetraethyl lead (TEL) is an anti-knocking compounds when mixed with petrol tend to improve the octane no. and therefore, decreases the knocking in the cylinder of the combustion engine.

125. (c) **4 — Dichlorination of propane forms four structural isomers (1,1-; 1,2-; 1,3-; and 2,2-dichloropropane).**

126. **(a) Cyclopropane** — Due to high ring strain, it undergoes ring-opening with bromine to give an open-chain product.

127. **(a) Reacts with water** — Grignard reagent is destroyed by water, so it must be prepared in dry ether.

128. (b) Petrol sample 30% *n*-heptane + 70% *iso*-octane since *iso*-octane is 70%. Hence, octane no. is 70.

129. (b) **Zn – Hg/HCl — Clemmensen reduction converts ketones to hydrocarbons (–CO– → –CH_2–).**

130. **(c) Sulphonation** — Heating with fuming H_2SO_4 (oleum) introduces –SO_3H group forming sulphonic acid.

131. (a) $\underset{\text{Propene}}{CH_3-CH=CH_2}+H_2\xrightarrow[300^oC]{Ni}\underset{\text{Propane}}{CH_3-CH_2-CH_3}$

132. (a) As the number of branches increases, surface area decreases, due to which Vander Waal forces of attraction decreases. Hence, boiling point also decreases.

133. (a) **Higher saturated hydrocarbon — Paraffin wax consists of long-chain alkanes (higher saturated hydrocarbons).**

134. **(b) Substitution reaction** — Chlorination of methane in UV light is a free-radical substitution reaction.

135. (b) $\underset{n-\text{butane}}{CH_3CH_2CH_2CH_3} \xrightarrow[\Delta]{\text{Anhyd. } AlCl_3} \underset{\text{iso butane}}{CH_3-\underset{\underset{CH_3}{|}}{CH}-CH_3}$

136. (c) $CH_3-CH_2-Br + \underset{\text{alk}}{KOH} \xrightarrow{\text{Dehydrohal ogenation}} CH_2=CH_2 + KBr + H_2O$

In alcoholic *KOH* alkoxide ions (RO^-) are present which is a strong base. They abstract proton from β-carbon of alkyl halide and favours elimination reaction

$$\underset{\text{Alcohol}}{ROH} + KOH \rightarrow \underset{\text{Potassium alkoxide}}{ROK} + H_2O$$

$$ROK \rightarrow \underset{\text{Alkoxide ion}}{RO^-} + K^+$$

$$RO^- + H-\overset{\beta}{C}H_2-\overset{\alpha}{C}H_2-Br \rightarrow ROH + CH_2=CH_2 + Br$$

137. (a) $Al_4C_3 + 6H_2O \rightarrow \underset{\text{Methane}}{3CH_4} + 2Al_2O_3$

138. (c) In C_2H_6, $C-C$ bond length is 1.54Å.

139. (a) $R-COOK + 2H_2O \xrightarrow{\text{Electrolysis}} \underset{\text{Alkane}}{R-R} + CO_2 + 2KOH + H_2$

140. (c) Gasoline — Gasoline (petrol) is a lighter fraction with lower boiling range than kerosene, diesel, and heavy oil, so it is obtained at the lowest temperature.